

① Differentiate between ethics, Law and Professional Responsibility with suitable example from any field.

Answer:

Differentiating among ethics, Law and Professional Responsibility with suitable example.

Ethics	Law	Professional Responsibility
Moral Principles about right or wrong behaviour Source is Personal Values and Social Norms Flexibility is broad and situation based	Rules set by government or higher up to legally enforce Source is Constitution, Statutes and regulations Specific and Normative	Duties defined by Professional Standards Source is Code of Conducts (workplace) Practical guideline within Professional

These three are necessary to exist

Computing Examples:

- **Ethics:** A developer refuse to build a hidden data tracking feature as it violates user Privacy.

• **Law:** A company must comply with data protection regulations.

• **Professional Responsibility:** An engineer follows security coding standards, documents limitations and discusses any two historical case of ethical failures in technology and explain the lesson learned from them.

Answer:

- 1) Facebook - Cambridge Analytical data misuse
A third Party App harvested personal data from millions of users and their friends,

It was an ethical failure because the users were not informed about how their data would be used. Profit and growth were prioritized over privacy.

Lesson learned was it must be informed about consent and it should be strictly visible to the user. Data collection must be minimized and controlled. There must be a platform for accountability. Privacy must be core priority.

(2) Therac-25 radiation Therapy accidents (1985-1987)

Software race conditions and inadequate safety interlocks in a radiation therapy machine led to massive overdose and severe injury and deaths.

It causes overreliance on software without independent hardware safeguards. Poor testing and inadequate handling of operator.

Lesson learned: Safety critical system need defense

defense in depth. Combine software checks with hardware interlocks and fail safe. There must be transparent incident response. And also that human factor matters.

⑥ Explain the principle of software engineering ethics, highlighting the issues related to professional responsibilities. Discuss how the ACM/IEEE code of ethics guides ethical decision-making.

Answer: The core ideas of software engineering ethics are;

- Public interest first (Protect, safety, privacy).
- Honesty and Transparency (Reports risks, limits).
- Quality and competence (Build reliable system).
- Professional judgment. (Avoid conflict of interest).
- Respect and fairness.
- Treats colleague and user ethically.

Professional responsibility issues :

- Keeping user data confidential and secure.
- Refusing harmful or deceptive features.
- Reporting bugs and vulnerabilities.

Following standards and laws.

- Considering social impact.

How the ACM/IEEE code guides decision:

- Provide 8 principles (Public, Client, Product, judgement, Management, Profession, colleagues, self)
- Use them as checklist.
- Helps justify the ethical actions.

⊗ Compare and contrast Utilitarianism and Deontology (Kantian ethics) as ethical frameworks.

How could each approach handle a software privacy dilemma?

Aspect	Utilitarianism	Deontology (Kantian)
Core idea	Choose the action that maximize overall happiness/benefit	Follow moral duties and rights regardless the outcome
Key thinkers	Jeremy Bentham	Immanuel Kant
Decision rule	Weight consequence for everyone affected	Apply universal rules
Flexibility	Flexible	Strict

Software Privacy dilemma example:

A company wants to analyze user private data to improve a health app.

Utilitarianism: Allow data use if it benefits clearly outweigh harms and risks. Strong safeguards required.

Deontology: Do not use private data without user consent whether it is good or bad.

⊙ Explain virtue ethics and describe how it can guide ethical decision making for software engineers.

Answer: virtue ethics follows and focuses on character rather than rules. It asks "what kind of person should I be?" instead of "what rules should I follow." The goal is to achieve specific virtue.

For an engineer, it means ethical behaviour is a professional behaviour. Virtues are:

(i) Integrity

(ii) Courage

(iii) Justice

(iv) Prudence

(v) Empathy.

The goal is to be a virtuous engineer that doesn't just write clean code. They write code that serves the well-being of the people using it.

Q Why is explainability a key ethical requirements for AI system used in high stakes domain like credit lending or criminal justice? Discuss two societal harms that can cause.

Answer: Explain why explainability is a key ethical requirement for AI system are

(1) Accountability: One can not fix errors or assign responsibility if they don't know why AI made a specific choice.

(2) Right to appeal: People have fundamental right to understand and challenge decisions that impact their freedom or finances.

Two social harms of "Black Box AI"

(1) Hidden discrimination: AI can secretly learn and scale historical biases. (Denying loan on based on demographic proxies) without developer noticing it.

(a) Erosion of Justice: Defendants cannot get a fair trial or defend themselves if they are unable to examine the hidden machine logic used to sentence them.

Ans. to the Ques. No. 07

Key principles of the ACM code of ethics.

(1) Contribute to the society.

(2) Avoid harmful Technology

(3) Be honest and trustworthy.

(4) Be fair and Respect privacy and confidentiality.

(5) Respect intellectual property.

Professional Codes are important because;

- Decision making framework

- To gain public trust.

- To achieve professional Protection.

Ans to the Ques No. 08.1)

Whistleblowing: It is the act of an employee exposing hidden information or activities within their organization that are illegal or unethical to the public.

Ethical Considerations:

- Public safety in contrast with employee loyalty.
- Exhausting internal options.
- Magnitude of the whistleblowing and sizing the risk.

Challenge for computing Professional:

- Career destruction
- Legal and Financial ruin
- Emotional Troll.

Ans. to the Ques No. 09

A realistic scenario:

A data scientist at a health tech company realize that they can build a highly accurate AI model to predict suicide risk or severe disease outbreak. To make the model break, they need to analyze users' private text message, real time location etc. without consent.

The conflict:

- (i) Contribute to society
- (ii) Respect Privacy.

How the ACM guides resolution:

- (i) Minimize data : The scientist should explore if they get the data with less invasive data.
- (ii) Informed data consent of the users.
- (iii) Holistic Risk Assessment and to show the ratio of good outcome and bad outcome.

Ans. to the Ques. No 10

Copyright:

It protects the actual and original expression of the work. In software, it protects the source code that someone wrote, the user interface graphics and written content. It prevents others from directly copy pasting my work.

Patents:

Patents is process that protects the underlying ideas, methods or functional inventions. It protects novel algorithm or unique technical process.

Trademarks:

It protects brand identity. In software, it protects the app name, logo, slogans. It prevents competitor to use confusingly similar branding to trick users.

Ans. to the Ques No. 11

Comparing open source licensing with proprietary licensing.

(i) Open source: The source code is made freely available. Anyone can legally view, modify and distribute it (Linux.)

(ii) Proprietary: The source code is legally hidden or restricted. Users purchase a license only to use the software. (Windows.)

Ethical Implications:

open source: It promotes transparency, global collaboration and equal access to technology.

The cons are it often lacks clear accountability if the software fails. It also can lead to exploitation.

Proprietary: It protects intellectual property and ensures creators are fairly compensated. The cons are it creates AI Black Box systems where users can't verify how their data is used.

Ans. to the Ques. No. 12.

Professional Decent: Expressing disagreement or concern about an unethical practice internally.

Whistleblowing: Escalating the issue internally to expose illegal or harmful activities after the internal channel ignored or had failed.

Key ethical Consideration:

- Severity of harm
- Internal Exhaustion

• Clear Evidence:

Potential Risk to Professional:

• Career destruction

• Legal and financial ruin.

• Personal toll.

Ans. to the Ques. No. 13.

The concept of informed consent:

Informed consent means a user must explicitly agree to their data being collected only after they fully understand what is being taken, why it is needed and who it will be shared with. It requires clear, plain language explanation, agreement

Ethical principle organization must follow:

- (i) Transparency of the data collection
- (ii) Data minimization
- (iii) Purpose limitation
- (iv) Security and full responsibility of data.
- (v) Controlling the user withdrawal.

Answer to the Ques No 14.

Ethical concern of surveillance:

- (1) Loss of privacy: Constant monitoring eliminates the fundamental human rights for private and anonymity.
- (2) The chilling Effect: When people know they are constantly watched, they self-censor themselves.
- (3) Bias and Discrimination to the algorithm biasness.

Example of misuse:

- Facial recognition at protest.
- Bossware in the workplace.
- Predictive Policing etc.

Ans. to the Ques. No. 15.

Copyright :

What it protects: The expression of an idea.

Best suited for: Protecting one's specific codebase from being directly copy-pasted, pirated or stolen.

Why: It is automatic, inexpensive and lasts a very long time. However, it doesn't stop a competitor from looking at your app or writing completely similar results by different code.

Patents:

What it protects: The underlying invention, method or functional process (The logic).

Best suited for: Groundbreaking algorithm, novel data compression method or tech process.

Why: It gives one strict monopoly on the idea itself for about 20 years. It prevents others from reverse engineering tools or functions.

Ans. to the Ques. No. 16

Algorithm Bias : It occurs when an AI system produce systematic repeatable errors that create unfair outcomes, often discriminating about specific group of people.

Cause of Biasness:

- Biased training data
- Flawed design
- Lack of representation.

Measure to ensure fairness:

- Audit the data.
- Continuous testing
- Diverse teams.

Ans. to the Ques. No. 17.

Importance of transparency and Explainability

(i) Building Trust: Users must understand how and why the AI makes decision that may impact their lives. (Medical Diagnosis)

(ii) Error Correction: Engineers cannot fix biases, security flaws or hallucination if AI operates as Black Box.

(iii) Legal Compliance: It ensures the system meets regulatory standards.

Who should be accountable:

(i) The organization;

(ii) The developers.

(iii) The operator/user.

Ans. to the Ques. No. 18.

The role of ethical leadership in IT:

Ethical leaders set the tone at the top. In the tech industry, where rapid innovation often outpace laws, leaders are primary defence against harmful products.

They are responsible for ensuring that business goal never overrides human well-being.

How leaders can Foster an Ethical Culture:

- Lead by example.

- Build psychological safety.
- Integrate ethics into SDLC (Software Development life cycle).
- Clean policies and channel.

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Ans. to the Ques: No. 19.

Differentiating among ethical hacking and cybercrime.

(i) Ethical hacking: Known as white hat, it is authorized testing of a system defence to find and fix vulnerability before they can be exploited.

(ii) Cyber Crime: Known as Black hat, it is unauthorized breaching of a system for malicious purpose such as financial gain, data theft etc.

Conditions for Ethical hacking are:

- Explicit permission
- Scope adherence
- Respect for privacy.
- Full disclosure.

Ans. to the Ques. No. 20.

Responsible Disclosure:

It is the ethical practice of privately reporting a security vulnerability to the software creator and giving them a set timeframe (90 days) to fix it before sharing the details with public. This protects user from immediate attacks while ensuring the flaw eventually gets passed.

Ethical guideline for security researcher:

- Never exploit the vulnerability while testing.
- Report privately only to the creator.
- Provide proper documentation proof.
- Respect the grace period.
- Inform the public.

Ans. to the Ques No. 21

Digital Divide:

It is the gap between the individuals, households or geographic areas that have access to modern information and communication technology and those that do not.

Social Implications:

(i) Educational inequality as students without reliable home internet or device fall behind in school.

(ii) Economic disadvantages, the severely attack economy.

(iii) Healthcare and service exclusion.

Measures to bridge the gap:

(i) Infrastructure Investment.

(ii) Affordability Program

(iii) Digital Literacy Training

(iv) Public access point.

Ans. to the Ques, No 22.

Green Computing:

It is an environmentally responsible practice of designing, manufacturing, using and disposing of computers, servers and other tech infrastructure.

The goal is to reduce carbon footprint, lower energy consumptions and minimize electronic waste.

How computing professional can contribute.

- Green software Engineering.

- Virtualization and Cloud computing.

- E-waste Reduction.

- Energy efficient Data centres.

- Efficient network infrastructure.

- Green procurement.

- Green IT policy.

- Green IT training.

Ans. to the Ques No. 23.

Conflict of Interest:

It occurs when an IT professional's interest such as financial ties or family gains clash with other professional duties, potentially biasing their judgement on loyalty to their employer.

Examples:

- **Vendor selection bias:** Recommending one company to purchase software or hardware from a specific vendor cause they got the commission.
- **Moonlighting:** Working a freelance coding job on the side for a direct competitor of one's primary employer, risking information leaks.

How to handle:

- (1) Disclose immediately.
- (2) Recuse themselves.
- (3) Strictly follow policy.

Ans. to the Ques No. 24.

Explaining the concept of justice and fairness according to the Islam core principle.

(i) Adl (Absolute Justice) : The requirements to give everyone their due right without discrimination.

(ii) Ihsan (Excellence / compassion) : Going beyond the bare minimum of the law to act with kindness and social responsibility.

(iii) Amanah (Trusteeship) : The belief that the resources, power and positions are trust from Allah and that must be managed ethically for the benefit of society.

Application on modern Professional Life:

(i) Fair Compensation

(ii) Equitable Hiring

(iii) Social Responsibility

(iv) Transparency

Ans. to the Ques. No. 2.5

Aristotle's Golden Mean is the idea that every virtue is precise balance between two opposite vices: a deficiency (too little) and an excess (too much). Being virtuous isn't about following a rigid rule, but about finding the middle ground that is appropriate for the specific situation.

It guides decision making by.

- (i) Identifying the extreme.
- (ii) Target the mean
- (iii) Context is the most important matters.
- (iv) Habitual Practice.

This guides the ethical decision making by encouraging professional by avoiding